

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO4: Implement machine learning techniques including decision tree algorithms, reinforcement learning, and build models for classification and decision-making. (BL3)

Assessment Tool Weightage: 50%

Duration : 1 hour

QUESTIONS: 4

Total Marks:40

Annexure A

ARTICLE REVIEW

Code of Conduct:

I N. Chawran Aryan (192365053) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

N. Chawran Aryan (192365053)

Signature of the student:

ARTICLE REVIEW

INSTRUCTIONS

Select any ONE peer-reviewed research article (published after 2019) from IEEE Xplore, Springer, Elsevier, or ACM Digital Library related to any of the following topics: *Inductive Learning, Decision Trees, Statistical Learning Methods (Naïve Bayes / Logistic Regression / SVM), or Reinforcement Learning applications.*

Submit: (i) Article citation details (ii) PDF or valid DOI link (iii) Written review as per the tasks below.

Task 1: Article Summary

- (a) Provide the full citation of the article (Author(s), Title, Journal/Conference, Year, DOI). State the domain/application area and the specific ML problem addressed.
- (b) Summarise the key objective of the article in your own words (150–200 words). Clearly state the research gap the authors identified and how they proposed to fill it.

Task 2: Methodology Analysis

- (a) Explain the ML technique(s) used in the article (e.g., Decision Tree variant, RL algorithm, Statistical model). Describe the dataset used—size, features, source, and how it was prepared.
- (b) Map the technique to CO 4 topics (Inductive Learning / Decision Trees / Statistical Learning / RL). Identify one methodological strength and one limitation in the authors' approach.

Task 3: Results & Critical Evaluation

- (a) Report the key results (accuracy, F1-Score, AUC-ROC, reward convergence, etc.) as presented by the authors. Create a comparative table if multiple models are benchmarked.
- (b) Critically evaluate the results: Are the reported metrics realistic? Are there potential biases (dataset, evaluation protocol)? What additional experiments would strengthen the findings?
- (c) Discuss real-world applicability: Where can this technique be deployed? What are the challenges in scaling the solution to industry (data availability, compute, ethical issues)?

Task 4: Reflection & Learning Outcome

- (a) State three key insights you gained from reading the article that deepen your understanding of CO 4 topics. Connect each insight to a specific concept from the course syllabus.
- (b) If you were to extend this research, propose one novel experiment or enhancement (different dataset / improved algorithm / new application domain). Justify its feasibility and expected impact.

RUBRICS FOR CALCULATION:

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Article	20%	Demonstrates clear and in-depth understanding of the article's purpose, concepts, and arguments	Shows good understanding with minor gaps	Partial understanding; misses key ideas	Lacks understanding of the article
Summary	20%	Provides concise and accurate summary highlighting all key points	Covers most key points with minor omissions	Incomplete or partially accurate summary	Poor or incorrect summary
Critical Analysis	25%	Provides insightful critique with strong reasoning and evaluation	Provides reasonable analysis with some justification	Superficial analysis; limited evaluation	No meaningful analysis
Use of Evidence	15%	Effectively uses examples, data, or references to support review	Uses some supporting evidence with minor gaps	Limited or weak supporting evidence	No supporting evidence provided
Organization	20%	Well-structured, clear, and coherent writing with proper language and flow	Generally clear with minor issues	Poor organization; lacks clarity	Disorganized and difficult to understand

ARTICLE REVIEW

Article Selected

Title: Predictive model and feature importance for early detection of type II diabetes mellitus

Authors: Eric Adua, Emmanuel Awuni Kolog, Ebenezer Afrifa-Yamoah, Bright Amankwah, Christian Obirikorang, Enoch Odame Anto, Emmanuel Acheampong, Wei Wang, and Antonia Yarney Tetteh

Journal: Translational Medicine Communications

Volume: 6

Article Number: 17

Year: 2021

DOI: 10.1186/s41231-021-00096-z

Publisher: Springer Nature

Domain: Healthcare / Medical Diagnosis

ML Problem: Early prediction and detection of Type II Diabetes Mellitus (T2DM) using machine learning classification algorithms.

The article was published on 23 August 2021 and compares four machine learning classifiers: Naïve Bayes, K-Nearest Neighbor, Support Vector Machine, and Decision Tree. The study also investigates the importance of different clinical features for predicting T2DM.

TASK 1: ARTICLE SUMMARY

(a) Citation Details, Domain and ML Problem

The selected research article is “**Predictive model and feature importance for early detection of type II diabetes mellitus**”, authored by Eric Adua, Emmanuel Awuni Kolog, Ebenezer Afrifa-Yamoah, Bright Amankwah, Christian Obirikorang, Enoch Odame Anto, Emmanuel Acheampong, Wei Wang, and Antonia Yarney Tetteh. It was published in *Translational Medicine Communications*, Volume 6, Article 17, in 2021.

DOI: 10.1186/s41231-021-00096-z

The research focuses on the **healthcare and medical diagnosis domain**. The specific machine learning problem is binary classification: predicting whether an individual has Type II Diabetes Mellitus or does not have Type II Diabetes Mellitus.

The researchers used clinical and biochemical attributes such as age, BMI, blood pressure, HbA1c, fasting blood sugar and lipid measurements. Four classification algorithms—Naïve Bayes, KNN, SVM and Decision Tree—were trained and compared.

The article is particularly relevant to CO4 because it demonstrates inductive learning, classification using Decision Trees, statistical/probabilistic learning through Naïve Bayes, and performance evaluation using metrics such as accuracy, precision, recall, F1-score and AUC-ROC.

(b) Key Objective and Research Gap

The main objective of the study is to investigate whether machine learning can be used to identify Type II Diabetes Mellitus at an early stage using clinical and biochemical information. Early detection is important because timely intervention can help reduce complications associated with diabetes. The researchers compared four machine learning algorithms—Naïve Bayes, K-Nearest Neighbor, Support Vector Machine and Decision Tree—to determine which model could best distinguish diabetic patients from healthy individuals. Another important objective was to identify the clinical features that contributed most strongly to diabetes prediction. The study used data from a Ghanaian population, thereby providing evidence from an African population rather than relying only on commonly used benchmark datasets. The authors also addressed the need for feature ranking by applying Recursive Feature Elimination. Their analysis identified HbA1c, total cholesterol and BMI as the three most important predictors. The research gap addressed by the study was the need for effective machine learning-based prediction and feature identification for T2DM in an African population. The proposed approach helps support early detection and potentially assists healthcare professionals in making informed decisions.

TASK 2: METHODOLOGY ANALYSIS

(a) ML Techniques and Dataset

The researchers used four machine learning classification algorithms:

1. **Naïve Bayes**
2. **K-Nearest Neighbor (KNN)**
3. **Support Vector Machine (SVM)**
4. **Decision Tree (DT)**

The purpose was to compare their ability to classify patients into two groups:

- 0 → Non-T2DM
- 1 → T2DM

Dataset

The dataset contained **438 participants**:

- 219 Type II Diabetes patients
- 219 healthy controls

Therefore, the dataset was balanced between the two classes.

The participants were recruited in Ghana. T2DM patients were recruited from Komfo Anokye Teaching Hospital, while healthy individuals were recruited from communities in the Kumasi metropolitan area.

The dataset contained **11 attributes**, including:

- Age
- BMI
- Systolic Blood Pressure (SBP)
- Diastolic Blood Pressure (DBP)

- HbA1c
- Fasting Blood Sugar (FBS)
- Total Cholesterol (TC)
- Triglycerides (TG)
- HDL-C
- LDL-C
- T2DM class/target

Data Preparation

The researchers performed several preprocessing operations.

1. Data cleaning

The data was examined for errors and outliers. Boxplot analysis was used and no outliers were detected.

2. Missing values

Only 17 data points were missing. The researchers used the **Expectation-Maximization (EM)** algorithm to estimate the missing values.

3. Scaling

The predictor variables were scaled to the range **0 to 1** because the features had different numerical ranges.

4. Feature selection

Recursive Feature Elimination (RFE) was used to rank the predictive importance of the attributes.

5. Train-test split

80% of the data was used for training and 20% for testing.

Therefore:

- Training = 350 participants
- Testing = 88 participants

The study reports 350 training observations and 68 testing observations after the class-balanced training construction; the paper's methodology should be followed as the source for the reported experimental setup.

(b) Mapping to CO4

The research can be mapped to the following CO4 concepts:

CO4 Concept Application in Article

Inductive Learning Models learn diabetes patterns from labelled patient data

Decision Trees Decision Tree used for T2DM classification

Statistical Learning Naïve Bayes used as a probabilistic classifier

Classification Patients classified as T2DM or non-T2DM

Feature Selection RFE used to identify important predictors

Model Evaluation Accuracy, Precision, Recall, F1-score and AUC used

Methodological Strength

A major strength is the **comparison of multiple machine learning algorithms** rather than relying on a single classifier. This allows the researchers to determine which technique performs best on the dataset.

Another strength is the use of **feature selection**, which identifies the most important clinical predictors and makes the results more informative.

Methodological Limitation

The dataset contained only **438 participants**, which is relatively small for developing a model intended for broad clinical deployment.

Another limitation is that the participants were obtained from a specific Ghanaian population. Therefore, the model may not generalize equally well to populations with different demographics, healthcare systems and disease characteristics.

TASK 3: RESULTS & CRITICAL EVALUATION

(a) Key Results

The study reported that all four classifiers performed above the stated 70% acceptable threshold for the evaluated classification measures.

The reported overall accuracy values were:

Algorithm	Accuracy	AUC-ROC
Naïve Bayes	Higher performance	0.87
SVM	84%	0.84
KNN	83%	0.85
Decision Tree	81%	0.81

Naïve Bayes was the best-performing classifier according to the study's overall evaluation, with an **AUC of 0.87**.

The study also reported that 82% of diabetic test data was detected by the Naïve Bayes classifier, of which 93% were accurately predicted.

The three most important features were:

1. **HbA1c**
2. **Total Cholesterol (TC)**
3. **BMI**

The least important features were reported as:

1. Age
2. HDL-C
3. LDL-C

These findings are directly reported by the authors.

(b) Critical Evaluation

The reported results indicate that machine learning can provide useful predictive performance for T2DM classification. The AUC values ranging from 0.81 to 0.87 indicate that all four models demonstrated good discrimination on the study dataset.

However, the results should not automatically be interpreted as proof that the models are ready for clinical

deployment.

Dataset Bias

The dataset was collected from a specific population in Ghana. A model trained on this population may not perform identically on Indian, European, American or other African populations.

Sample Size

The dataset contains only 438 participants. A larger and more diverse dataset would provide stronger evidence of generalization.

Sampling

The researchers used purposive sampling for T2DM patients and convenience sampling for healthy controls. This may introduce selection bias.

Evaluation

The study used an 80:20 train-test division. A stronger evaluation could include:

- Repeated k-fold cross-validation
- External validation dataset
- Independent hospital validation
- Confidence intervals for performance metrics
- Calibration analysis
- Subgroup analysis

Additional Experiments

The study could be strengthened by testing the models on:

- Larger datasets
- Multi-hospital datasets
- Different ethnic populations
- External validation datasets
- Longitudinal patient records

The authors themselves emphasize the potential of these models for early detection, but broader validation would be necessary before real clinical deployment.

(c) Real-World Applicability

The proposed machine learning approach could be used as a **clinical decision-support system**.

Potential applications include:

1. Hospital Screening

Patient information could be entered into the system and the model could estimate the likelihood of T2DM.

2. Primary Healthcare

The system could help healthcare workers identify patients who require additional testing.

3. Community Screening

A lightweight prediction model could support diabetes screening programs in communities with limited medical resources.

4. Electronic Health Records

The model could potentially be integrated with electronic health record systems to identify high-risk

patients.

Industry Challenges

Several challenges must be addressed before deployment:

- Availability of large, representative datasets
- Patient privacy
- Data security
- Model bias
- Regulatory requirements
- Clinical validation
- Explainability
- Integration with existing hospital systems
- Continuous model monitoring

The model should therefore be treated as a **decision-support tool rather than an autonomous diagnostic system**.

TASK 4: REFLECTION & LEARNING OUTCOME

(a) Three Key Insights

Insight 1: Inductive Learning

I learned that machine learning classification is an example of **inductive learning**, where a model learns patterns from labelled examples and uses those patterns to classify previously unseen patients.

This connects directly with the CO4 concept of **classification and inductive learning**.

Insight 2: Decision Tree Classification

The article helped me understand how a Decision Tree can convert medical attributes into a sequence of decisions for predicting diabetes.

The Decision Tree achieved an AUC of **0.81**, demonstrating that it was able to distinguish diabetic and non-diabetic patients.

This connects directly with the CO4 topic of **Decision Tree algorithms**.

Insight 3: Importance of Feature Selection

The study demonstrated that not every feature contributes equally to prediction. RFE was used to rank the predictors, and HbA1c, total cholesterol and BMI were identified as the most important attributes.

This connects to **feature selection and model optimization**, which are important parts of machine learning model development.

(b) Proposed Novel Experiment

If I were to extend this research, I would develop a **multi-population diabetes prediction model using federated learning**.

Instead of collecting all patient data in one central database, hospitals could train the model locally using their own patient records. Only model updates would be shared between institutions.

Proposed Experiment

The experiment would involve:

1. Collecting diabetes records from multiple hospitals.
2. Keeping patient data within each hospital.

3. Training the same machine learning model locally.
4. Combining model updates using federated learning.
5. Comparing the federated model with a centrally trained model.
6. Evaluating Accuracy, Precision, Recall, F1-score and AUC-ROC.
7. Performing subgroup fairness analysis.

Expected Impact

This approach could improve:

- Data privacy
- Dataset diversity
- Generalization
- Model robustness
- Collaboration between hospitals

It would also address one of the major limitations of the original research: the relatively small and geographically limited dataset.

CONCLUSION

The reviewed article demonstrates that machine learning can be useful for the early prediction of Type II Diabetes Mellitus. The researchers compared Naïve Bayes, KNN, SVM and Decision Tree algorithms and found that Naïve Bayes produced the strongest overall performance, with an AUC of 0.87. The study also identified HbA1c, total cholesterol and BMI as the most important predictive features.

From a CO4 perspective, the article provides practical understanding of **inductive learning, classification, Decision Trees, statistical learning, feature selection and model evaluation**. However, the relatively small dataset and population-specific sampling mean that additional external validation is required before clinical deployment.

Overall, the research shows how machine learning can support early diabetes detection and healthcare decision-making while also demonstrating the importance of data quality, model evaluation, feature selection, fairness and clinical validation.